

---

# LatentAudit: Resource-Efficient White-Box Auditing for Trustworthy RAG in High-Stakes Domains

---

Anonymous Author(s)

Affiliation

Address

email

## Abstract

1 Retrieval-augmented generation can retrieve the right source and still produce an  
2 answer that departs from it. Checking every answer with a second LLM adds  
3 latency, cost, and possible data egress. We ask whether the generator’s own  
4 activations contain a cheaper warning signal. *LatentAudit* aligns frozen evidence  
5 embeddings with pooled answer states on faithful calibration pairs, then scores  
6 covariance-normalized alignment residuals. The rule is fit in closed form and adds  
7 0.24–0.77 ms of post-generation computation in our measurements. Across three  
8 QA domains and five open-weight model families, this signal approaches GPT-4o  
9 judging and transfers to HaluEval. RAGTruth exposes its boundary: zero-shot  
10 AUROC is 0.6947 and rises to 0.7816 after 200-example recalibration because  
11 whole-answer pooling can dilute localized errors. LatentAudit is therefore a first-  
12 pass routing monitor, not a truth oracle or claim localizer. It requires open weights,  
13 trustworthy retrieval, and validation after deployment shifts.

## 14 1 Introduction

15 RAG systems increasingly mediate access to biomedical, legal, and public-service information.  
16 Retrieval narrows the source base, but it does not force the answer to follow that source. Institutions  
17 therefore need to catch evidence departures without sending every sensitive query to another large  
18 model.

19 The usual checks create an operational trade-off. LLM judges [Zheng et al., 2023] require another  
20 inference call, while sampling-based methods [Manakul et al., 2023] generate several alternatives.  
21 Both add latency and compute; hosted judges can also expose the retrieved context. Internal activations  
22 offer a local alternative, yet generic factuality signals do not measure whether an answer follows a  
23 particular passage.

24 This leaves a narrower question: after aligning evidence and answer representations, do faithful  
25 generations exhibit a recognizable *evidence-residual signal*? Prior work shows that internal activations  
26 track aspects of factual recall and truthful generation [Meng et al., 2022, Li et al., 2023b]. We test  
27 whether deviations from the residual pattern of faithful evidence–answer pairs can serve as an  
28 inexpensive warning.

29 LatentAudit implements this test with closed-form calibration. It pools answer-token activations,  
30 maps a frozen evidence embedding into the answer-state space, and estimates the covariance of  
31 faithful alignment residuals. A Mahalanobis score then flags directions that are unusual for calibrated  
32 faithful pairs. The method trains no auxiliary neural classifier and needs no second generative judge.

33 Our evaluation moves from controlled separation to external benchmarks and explicit failure cases.  
34 We test three QA domains, five model families, harder retrieval failures, HaluEval, RAGTruth,  
35 mixed-claim responses, and a 32B generator. The controlled settings show strong separation at

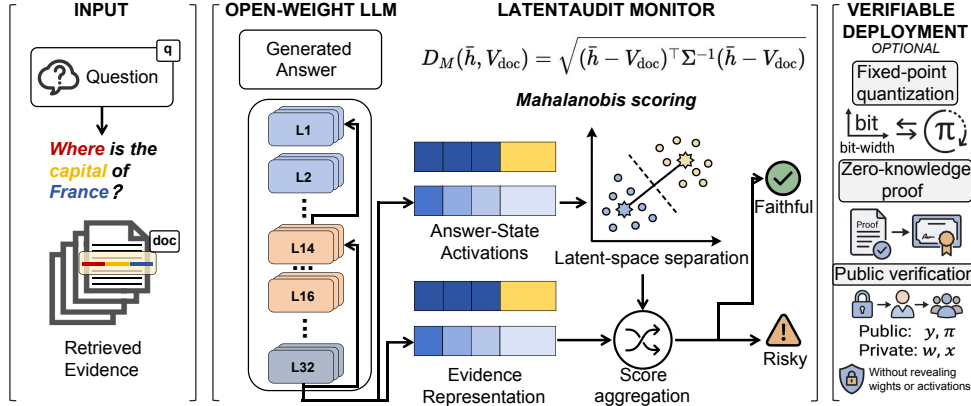


Figure 1: LatentAudit reuses hidden states from an open-weight RAG generation, compares an answer-state summary with the retrieved evidence, and returns a calibrated risk flag. The optional verification path attests execution of the registered score; it does not prove that the answer or evidence is true.

sub-millisecond incremental cost. RAGTruth shows where the abstraction breaks: a response-level centroid can dilute a short unsupported span.

We organize the paper around three questions:

1. **RQ1: Does the evidence-residual signal support a lightweight alarm?** We measure discrimination and incremental cost.
2. **RQ2: Where does the alarm transfer or fail?** We test model, domain, scale, retrieval, external-benchmark, and mixed-claim shifts.
3. **RQ3: What does responsible use require?** We examine local execution, calibration, reproducibility, and limits on truth, privacy, and model access.

## 2 Related Work

**Internal signals for factuality.** Causal tracing [Meng et al., 2022] and inference-time intervention [Li et al., 2023b] identify activations associated with factual recall and truthful generation. Evidence support is a different relation: it asks whether this answer follows this passage. LatentAudit tests whether cross-space alignment makes that relation useful for monitoring.

**Faithfulness verification.** LLM judges [Zheng et al., 2023] and sampling-based checks [Manakul et al., 2023] are flexible but add inference. SynCheck combines likelihood, uncertainty, context influence, and semantic alignment during RAG generation [Wu et al., 2024]. LatentAudit targets a local, evidence-conditioned score with closed-form calibration. Our SynCheck comparison is partial because cosine replaces AlignScore.

**Verifiable computation.** EZKL compiles supported numerical computation graphs into zero-knowledge circuits [zkonduit, 2022]. We use this only as an optional execution-attestation path. A proof can certify that the registered score was evaluated; it cannot certify that retrieved evidence is true or that the monitor is empirically valid.

## 3 Methodology

LatentAudit has one core component and one optional deployment component (Figure 1). The core monitor reuses states from generation. The verification path, when required, wraps only the monitor’s output in a proof.

### 63 Answer-State Representation

64 Let  $\mathcal{X} = [x_1, \dots, x_N]$  concatenate the question and retrieved context  $\mathcal{C}$ . During decoding, layer  $\ell$   
 65 produces token states  $h_\ell^{(t)} \in \mathbb{R}^d$ . We select the audit layer  $\ell^*$  on the calibration split, then mean-pool  
 66 the most salient answer-token states into  $V_{act}$ . Qwen2.5-7B, Llama-3-8B, and Qwen2.5-32B select  
 67 L14/28, L16/32, and L32/64, respectively.

### 68 Cross-Space Residual Geometry and Decision Rule

69 The answer state and evidence embedding come from different encoders. We obtain an evidence  
 70 vector  $e$  from the frozen retriever all-MiniLM-L6-v2, then align it with the answer-state space. On  
 71 faithful calibration pairs  $(e_i, a_i)$ , ridge regression fits

$$W_{\text{proj}} = \arg \min_W \sum_{i \in \mathcal{S}_{\text{calib}}^+} \|a_i - W e_i\|_2^2 + \lambda \|W\|_F^2. \quad (1)$$

72 This is supervised, closed-form calibration; it does not assume that MiniLM and the generator share  
 73 an encoder or latent manifold. The appendix reports projector-complexity and sample-efficiency  
 74 ablations.

75 For each faithful pair, define the alignment residual  $r_i = a_i - W_{\text{proj}} e_i$ . LatentAudit estimates the  
 76 covariance of these residuals with Ledoit–Wolf shrinkage. At test time, it asks whether a new  
 77 answer–evidence residual is typical of the faithful calibration set.

78 Because high-dimensional LLM representations are anisotropic, Euclidean distance can overempha-  
 79 size common directions. We instead use a Mahalanobis score with the estimated precision matrix  
 80  $\Sigma^{-1}$ :

$$D_M(V_{act}, e) = \sqrt{(V_{act} - W_{\text{proj}} e)^\top \Sigma^{-1} (V_{act} - W_{\text{proj}} e)}. \quad (2)$$

81 Unsupported answers can remain close in cosine space through topical overlap, yet deviate along  
 82 directions rarely observed for faithful pairs. The threshold  $\tau^*$  maximizes Youden’s J on the calibration  
 83 split. Scores above  $\tau^*$  are routed as potentially unsupported.

### 84 Optional Execution Attestation

85 The squared distance is a quadratic form, so a fixed-point implementation can be compiled into  
 86 an arithmetic circuit with EZKL [zkonduit, 2022]; succinct pairing-based arguments provide one  
 87 verification construction [Groth, 2016]. This optional path binds a score to a registered threshold and  
 88 input hash without publishing the latent vectors. It certifies execution of that rule only; circuit details  
 89 and the explicit scope of the claim appear in the appendix.

## 90 4 Experiments and Results

### 91 Experimental Setup and Baselines

92 We evaluate on PubMedQA [Jin et al., 2019], HotpotQA [Yang et al., 2018], and TriviaQA [Joshi  
 93 et al., 2017]. They cover biomedical QA, Wikipedia-based multi-hop QA, and open-domain reading  
 94 comprehension. For each domain, we construct 2,000 generations balanced between faithful and  
 95 induced unsupported cases. We create the latter by replacing the retrieved context with an adversarial  
 96 contradiction while holding the question and generation pipeline fixed. A 200-example calibration  
 97 split selects the layer and fits  $W_{\text{proj}}$ ,  $\Sigma^{-1}$ , and  $\tau^*$ . All parameters are then frozen for the disjoint  
 98 1,800-example evaluation split.

99 For PubMedQA and HotpotQA, we also build four-way stress tests from 400 source questions. Each  
 100 source yields faithful, contradicted, retrieval-miss, and partial-support variants. Retrieval misses use  
 101 topically similar but source-mismatched evidence. Partial-support cases retain relevant context but  
 102 remove information needed to support the full answer. We then evaluate on 6,000 HaluEval QA pairs  
 103 [Li et al., 2023a] and the 2,675-response RAGTruth test set [Niu et al., 2024]. HaluEval combines  
 104 generated-and-filtered hallucinations with human-annotated outputs. RAGTruth contains naturally  
 105 generated RAG responses with localized span annotations.

106 We benchmark *LatentAudit* against five detection methods spanning four paradigms:

- **LLM-as-a-Judge (GPT-4o):** A reference-based zero-shot judge that receives the question, evidence, and candidate answer and returns a binary supported/unsupported verdict at  $T=0.0$ .
- **SelfCheckGPT [Manakul et al., 2023]:** Estimates inconsistency from  $N=10$  independent generations at  $T=1.0$ .
- **INSIDE [Chen et al., 2024] & SAPLMA [Azaria and Mitchell, 2023]:** INSIDE defines EigenScore from covariance eigenvalues of internal response embeddings across sampled generations; SAPLMA learns a shallow classifier from hidden activations. For the common-split comparison, we instantiate their published signal types as calibrated detectors on the same split as LatentAudit. These are controlled adaptations, not official reproductions.
- **SynCheck [Wu et al., 2024]:** Combines likelihood, entropy/LID, context-influence, and semantic-alignment features. Our controlled source-grouped comparison reproduces its feature pipeline, substituting cosine for AlignScore as explicitly noted below.
- **Perplexity-Based (Min- $\mathbb{P}$ ):** Classifies hallucination from token log-likelihood statistics over the generated sequence.

The main-text tables report evaluations across five audited model families spanning Llama-2, Llama-3 [Team, 2024], Qwen2.5 [Yang et al., 2024], Qwen3 [Team, 2025], and Mistral [Jiang et al., 2023], all executed at FP16 precision. Unless otherwise stated, summary-table variation is computed from five evaluation-set resamples. The separate threshold-stability analysis uses 200 calibration bootstraps.

**Reproducibility Notes.** For the GPT-4o baseline, each evaluation instance is serialized as (question, retrieved evidence, candidate answer) and scored with a binary instruction of the form: “Is the answer fully supported by the evidence? Reply with SUPPORTED or UNSUPPORTED.” For SelfCheckGPT, we follow the original repeated-sampling recipe and compare the candidate answer against  $N = 10$  stochastic generations produced under the same question and retrieved context. For LatentAudit, all covariance and threshold parameters are fit only on the calibration split and then frozen before evaluation. The main benchmark, cross-model table, OOD study, and stress-test results all reuse this protocol.

#### RQ1: Does the evidence-residual signal support a lightweight alarm?

Table 1 compares detection quality and incremental latency. GPT-4o has the highest AUROC but takes more than five seconds per query in our setup. LatentAudit trails it by 0.6 AUROC points and 1.2 F1 points on Llama-3-8B while scoring in under one millisecond. The controlled INSIDE- and SAPLMA-style baselines fall between these two operating points.

Table 1: RQ1: a single Mahalanobis monitor closes most of the gap to GPT-4o while remaining sub-millisecond. Latency is measured per query; “proving” refers to the optional ZK layer.

| Method                    | Latency (ms)                | Llama-3-8B   |              | Qwen-2.5-7B  |              | Mistral-7B   |              |
|---------------------------|-----------------------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                           |                             | AUROC        | F1           | AUROC        | F1           | AUROC        | F1           |
| GPT-4o Judge              | ~5,300                      | 0.948        | 0.881        | 0.945        | 0.876        | 0.940        | 0.870        |
| SelfCheckGPT              | ~28,500                     | 0.871        | 0.804        | 0.865        | 0.798        | 0.858        | 0.790        |
| INSIDE-style <sup>†</sup> | ~3.8                        | 0.908        | 0.841        | 0.901        | 0.832        | 0.895        | 0.825        |
| SAPLMA-style <sup>†</sup> | ~1.5                        | 0.882        | 0.815        | 0.876        | 0.808        | 0.870        | 0.800        |
| Min-Perplexity            | 0.0                         | 0.722        | 0.655        | 0.718        | 0.650        | 0.710        | 0.642        |
| <b>LatentAudit (Ours)</b> | <b>0.77 (+11.2 proving)</b> | <b>0.942</b> | <b>0.869</b> | <b>0.938</b> | <b>0.862</b> | <b>0.925</b> | <b>0.852</b> |

<sup>†</sup>Controlled common-split adaptations of the published signal types; not official reproductions.

**Latency scope.** The 0.77 ms entry measures only the audit added to an existing RAG generation. Hidden states come from the generation pass, and the evidence embedding is cached at retrieval time. On Qwen2.5-32B, extraction, pooling, projection, scoring, and thresholding take 0.24 ms on one A100. The shared model forward pass takes 111 ms; the appendix gives the full profile.

Pooling is the main representation choice behind Table 1. On Llama-3-8B and PubMedQA, mean-pooling the eight most salient tokens reaches 0.942 AUROC. Last-token and max-pooling alternatives

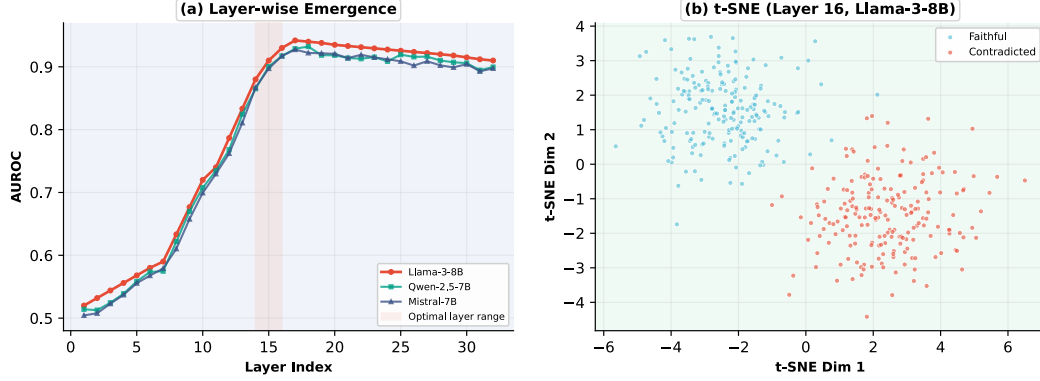


Figure 2: RQ1 diagnostic: the best discrimination occurs in the mid-to-late residual stream. The reported optima are L14/28, L16/32, and L32/64 for Qwen2.5-7B, Llama-3-8B, and Qwen2.5-32B, respectively.

reach 0.884 and 0.912. The gain motivates a response centroid, while the mixed-claim study later shows its cost in localization.

The affine projector is also necessary. Table 2 shows that Ridge improves evaluation AUROC from 0.648 without projection to 0.942. A two-layer MLP reaches 0.945, but its train–evaluation gap is much larger. We retain Ridge because it is closed-form and comes within 0.003 AUROC of the nonlinear alternative.

Table 2: RQ1 ablation: supervised alignment accounts for most of the gain. Ridge nearly matches the nonlinear MLP while preserving closed-form calibration.

| Alignment           | Train AUROC ( $N=200$ ) | Eval AUROC ( $N=1,800$ ) |
|---------------------|-------------------------|--------------------------|
| No projection       | 0.654                   | 0.648                    |
| PCA (unsupervised)  | 0.785                   | 0.778                    |
| CCA                 | 0.892                   | 0.885                    |
| <b>Ridge (ours)</b> | <b>0.948</b>            | <b>0.942</b>             |
| MLP (two layer)     | 0.991                   | 0.945                    |

**Where does the signal emerge?** Figure 2(a) sweeps layers in three models. Qwen2.5-7B, Llama-3-8B, and Qwen2.5-32B peak at L14/28, L16/32, and L32/64. These mid-depth optima are consistent with prior evidence that factuality-related activations can be localized inside transformers [Meng et al., 2022, Li et al., 2023b]. They do not establish a shared mechanism. For each new architecture, we select the audit layer only on its calibration split. Figure 2(b) visualizes the resulting separation.

## RQ2: Where does the alarm generalize or fail?

Table 3 tests transfer across backbones and domains. AUROC ranges from 0.925 to 0.948 on PubMedQA and from 0.915 to 0.940 on TriviaQA. HotpotQA is harder, ranging from 0.905 to 0.928. The signal therefore persists across the five tested model families, with a consistent drop on multi-hop questions.

**External hallucination benchmarks and 32B scaling.** Table 4 summarizes the external and scale checks. HaluEval transfers without target-set recalibration. RAGTruth is harder: its errors can occupy short spans inside otherwise supported answers, and Qwen2.5-7B audits outputs from several generators. Its zero-shot AUROC is 0.6947; 200-example recalibration raises it to 0.7816. The repeated-split result uses a separate bundle protocol and is not combined with the recalibrated result. The 32B row extends scale within a controlled PubMedQA stress setting rather than a new natural benchmark.

Table 3: RQ2a: the same calibration procedure yields a useful signal across the tested model families and domains.

| Dataset Domain         | Llama-2 (7B)     | Llama-3 (8B)     | Qwen-2.5 (7B)    | Qwen-3 (8B)      | Mistral (7B)     |
|------------------------|------------------|------------------|------------------|------------------|------------------|
| PubMedQA (Medical)     | $0.931 \pm 0.02$ | $0.942 \pm 0.01$ | $0.938 \pm 0.02$ | $0.948 \pm 0.01$ | $0.925 \pm 0.01$ |
| TriviaQA (Open-domain) | $0.915 \pm 0.02$ | $0.935 \pm 0.01$ | $0.928 \pm 0.02$ | $0.940 \pm 0.01$ | $0.918 \pm 0.02$ |
| HotpotQA (Multi-hop)   | $0.905 \pm 0.02$ | $0.928 \pm 0.01$ | $0.918 \pm 0.02$ | $0.922 \pm 0.02$ | $0.910 \pm 0.02$ |

Table 4: RQ2 external and scale checks. Rows use distinct protocols and are reported separately rather than averaged.

| Evaluation            | Protocol                  | N     | AUROC             | F1     |
|-----------------------|---------------------------|-------|-------------------|--------|
| HaluEval QA           | Zero-shot transfer        | 6,000 | 0.9496            | —      |
| RAGTruth              | Zero-shot transfer        | 2,675 | 0.6947            | —      |
| RAGTruth              | 200-example recalibration | 2,675 | 0.7816            | 0.8124 |
| RAGTruth              | Repeated-split bundle     | 2,675 | $0.788 \pm 0.001$ | —      |
| PubMedQA stress (32B) | Calibrated bundle         | 1,600 | 0.9888            | 0.8765 |

**SynCheck comparison.** On 800 PubMedQA stress examples grouped by 200 source questions, the ten-feature SynCheck reproduction reaches 0.9925 AUROC. LatentAudit’s geometric/LID bundle reaches 0.9938 with one forward pass. Across five source-level splits, their means are 0.9829 and 0.9922, respectively. Because cosine replaces SynCheck’s AlignScore component, this is a controlled partial reproduction, not an official re-evaluation.

**Mixed-claim evaluation.** We next interleave five faithful claims with unsupported claims at three levels of topical proximity. Faithful claims have mean  $D_M = 3.12$ ; injected claims range from 6.84 to 12.05. Some domain-consistent errors have higher raw cosine than faithful claims, while the covariance-adjusted score separates every injected case. This small controlled study tests claim-scoped scoring. RAGTruth provides the complementary natural span setting, where response pooling remains much harder.

Table 5: RQ2 controlled claim-scope diagnostic. Topical similarity can remain high for unsupported claims, whereas the covariance-adjusted residual increases. The five-claim faithful row is an aggregate; this is not a natural claim-level benchmark.

| Claim type                          | Raw cosine | $D_M$ |
|-------------------------------------|------------|-------|
| Faithful claims ( $n=5$ , mean)     | 0.7768     | 3.12  |
| Domain-consistent: tumor regression | 0.8112     | 6.84  |
| Domain-consistent: metastatic cure  | 0.7730     | 7.15  |
| Off-topic medical: sleep quality    | 0.7577     | 8.42  |
| Off-topic medical: hay fever        | 0.8015     | 7.96  |
| Extreme off-topic: basketball       | 0.7523     | 11.23 |
| Extreme off-topic: weather          | 0.7342     | 12.05 |

**Within-family calibration transfer.** An isolated five-seed transfer study between Llama-2-7B and Llama-3-8B uses 800 PubMedQA examples and the raw single-feature Mahalanobis score. Transferring  $W_{\text{proj}}$  changes AUROC from  $0.6561 \pm 0.0121$  to  $0.6533 \pm 0.0220$  ( $-0.003$ ), whereas transferring  $\Sigma^{-1}$  yields  $0.5897 \pm 0.0313$  ( $-0.066$ ). Thus the evidence projector is nearly conserved, while covariance retains a measurable model-specific component; these values should not be compared directly with the full bundle’s 0.94+ AUROC.

**Across model families.** Figure 3 shows the calibrated score distributions. Faithful and contradicted interquartile ranges do not overlap in the plotted model–domain pairs. The appendix further disaggregates these distributions by model; thresholds remain model-specific unless explicitly transferred.

**Beyond outright contradictions.** Retrieval failures are more varied than outright contradictions. We approximate this variation with the controlled four-way corpus in Section 4. Across four model

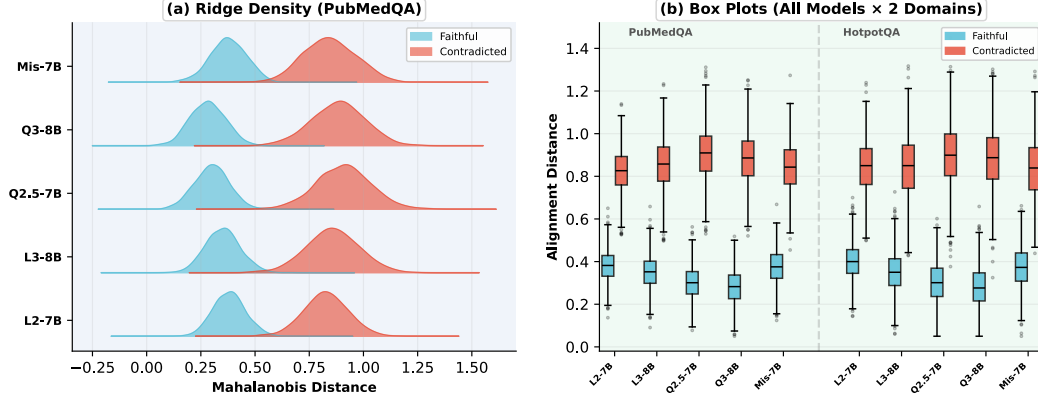


Figure 3: RQ2 diagnostic: faithful and contradicted distributions remain separated after per-model calibration across the tested families and domains.

Table 6: RQ2b: in controlled retrieval failures, partial-support examples are the hardest negatives. Abbreviations: F/C = faithful vs. contradicted, F/RM = faithful vs. retrieval-miss, F/P = faithful vs. partial.

| Domain   | Model    | AUROC $\uparrow$ | AUPRC $\uparrow$ | F1 $\uparrow$ | F/C    | F/RM   | F/P    |
|----------|----------|------------------|------------------|---------------|--------|--------|--------|
| PubMedQA | Llama-2  | 0.9776           | 0.9387           | 0.8322        | 0.9971 | 0.9667 | 0.9727 |
|          | Llama-3  | 0.9815           | 0.9450           | 0.8510        | 0.9982 | 0.9710 | 0.9755 |
|          | Qwen-2.5 | 0.9566           | 0.8806           | 0.8025        | 0.9938 | 0.9542 | 0.9218 |
|          | Qwen-3   | 0.9682           | 0.9102           | 0.8244        | 0.9950 | 0.9622 | 0.9445 |
| HotpotQA | Llama-2  | 0.9142           | 0.7760           | 0.7312        | 0.9880 | 0.9688 | 0.8205 |
|          | Llama-3  | 0.9315           | 0.8214           | 0.7855        | 0.9925 | 0.9760 | 0.8550 |
|          | Qwen-2.5 | 0.9207           | 0.7698           | 0.7575        | 0.9863 | 0.9653 | 0.8364 |
|          | Qwen-3   | 0.9280           | 0.8045           | 0.7780        | 0.9895 | 0.9720 | 0.8410 |

families, PubMedQA AUROC ranges from 0.9566 to 0.9815; HotpotQA ranges from 0.9142 to 0.9315.

The pairwise columns locate the difficulty. Contradicted and retrieval-miss variants separate more cleanly than partial-support variants. Partial support retains topical overlap while removing required evidence, so relevance alone cannot settle the label. Pairwise AUROC remains 0.9218 on PubMedQA but falls to 0.8364 on HotpotQA.

A 100-seed Llama-3-8B spot check on PubMedQA yields 0.9833 AUROC, extending the stress-test result beyond Qwen.

Across 200 calibration bootstrap resamples, the PubMedQA threshold has  $\sigma=0.063$  and test F1 has  $\sigma=0.019$ . On HotpotQA, the corresponding values are 0.086 and 0.024.

Errors concentrate in interpretable cases. PubMedQA false positives often retain high lexical overlap but insufficient evidence. HotpotQA false negatives often have single-token answers, which provide a weak state summary. Evidence incompleteness under topical overlap is therefore the main observed failure mode.

**Without target-domain recalibration.** A practical deployment may not have labeled data from every target domain. We therefore fit  $W_{\text{proj}}$ ,  $\Sigma^{-1}$ , and  $\tau^*$  on PubMedQA and apply all three without modification to HotpotQA, and vice versa.

Table 7 shows a 2–3 point AUROC drop in each direction. The signal transfers partially, but the acceptable loss depends on the deployment’s review cost and error tolerance.

Table 7: RQ2c: cross-domain calibration transfer reduces AUROC by 2–3 points.

| Calibration Domain   | Evaluation Domain    | In-Domain AUROC | OOD AUROC |
|----------------------|----------------------|-----------------|-----------|
| PubMedQA (Medical)   | HotpotQA (Multi-hop) | 0.942           | 0.916     |
| HotpotQA (Multi-hop) | PubMedQA (Medical)   | 0.928           | 0.902     |

### 209 RQ3: What does responsible use require?

210 LatentAudit runs locally, reuses generation states, and exposes its calibration rule. The reported  
 211 0.77 ms is incremental audit time, not end-to-end RAG latency. A separate Qwen2.5-32B profile  
 212 measures 0.24 ms on one A100, compared with a 111 ms shared forward pass. For later attestation,  
 213 16-bit fixed-point evaluation retains 99.8% of FP16 AUROC in our experiment. The appendix reports  
 214 the quantization and proof details. These properties make the score easier to operate and audit; they  
 215 do not make it correct by construction.

216 **Calibration burden and runtime.** Table 8(a) shows that evaluation AUROC rises from 0.912  
 217 with 50 faithful calibration pairs to 0.942 with 200, then plateaus. Panel (b) separates the reused  
 218 generation work from the added audit. Extraction and pooling cost 0.14 ms; projection, scoring, and  
 219 thresholding add 0.10 ms. The 0.24 ms total therefore excludes both the shared generator pass and  
 220 the evidence embedding cached at retrieval time.

Table 8: RQ3 operational profile. (a) On Llama-3-8B/PubMedQA, the default 200-pair calibration lies near the observed plateau. (b) On Qwen2.5-32B, the audit adds 0.24 ms after generation; shared and cached work is shown only for scope.

| (a) Calibration efficiency |              | (b) Execution scope         |              |
|----------------------------|--------------|-----------------------------|--------------|
| Faithful pairs             | Eval AUROC   | Component                   | Latency (ms) |
| 50                         | 0.912        | RAG forward (shared)        | ~111         |
| 100                        | 0.931        | Evidence embedding (cached) | ~106         |
| <b>200</b>                 | <b>0.942</b> | Extraction + pooling        | 0.14         |
| 500                        | 0.943        | Projection + decision       | 0.10         |
| 1,000                      | 0.943        | <b>Incremental audit</b>    | <b>0.24</b>  |

221 **Concrete error behavior.** Table 9 makes the stress-test boundary tangible. Contradictions and  
 222 retrieval misses receive large distances. Partial support is harder: relevant context can still omit the  
 223 fact needed to justify the answer. This is the same failure pattern seen in the aggregate F/P column of  
 224 Table 6.

Table 9: RQ3 controlled PubMedQA examples (Llama-3-8B;  $\tau^* \approx 5.4$ ). These examples illustrate decision behavior rather than estimate error frequency.

| Condition       | Evidence                                  | Answer                     | $D_M$ | Decision    |
|-----------------|-------------------------------------------|----------------------------|-------|-------------|
| Faithful        | “therapy significantly reduced mortality” | Therapy reduces mortality. | 3.2   | Pass        |
| Contradicted    | “therapy had no effect on mortality”      | Therapy reduces mortality. | 7.8   | Reject      |
| Retrieval miss  | “patients received placebo”               | Therapy reduces mortality. | 8.5   | Reject      |
| Partial support | “therapy was evaluated in 100 patients”   | Therapy reduces mortality. | 5.9   | Reject (FP) |

## 225 5 Public-Interest Deployment and Limitations

226 LatentAudit is a first-pass warning system for open-weight RAG, not an autonomous arbiter. It avoids  
 227 a second generative call and can keep retrieved context and activations inside the deploying institution.  
 228 This operating point may help organizations with tight compute or data-governance constraints. Our

experiments do not establish safety in hospitals, legal services, libraries, or public agencies without local validation.

**What the geometry measures.** The projector learns the typical relation between frozen evidence embeddings and faithful answer states. The covariance then records which alignment residuals are common in the calibration set. A high score therefore marks an unusual cross-space residual, not a causal explanation of hallucination.

**Open weights required.** The monitor reads  $h_{\ell^*}^{(t)}$  and therefore cannot directly audit a black-box API. A surrogate would measure its own representation rather than the original generator’s state; we leave that setting unvalidated.

**Quantization noise near the boundary.** Mapping floating-point activations to  $\mathbb{F}_p$  via  $\hat{V} = \text{round}(V \cdot 2^k)$  introduces rounding error. Samples whose true  $D_M$  falls near  $\tau^*$  may cross the boundary after quantization. We mitigate this with a continuous safety margin (detailed in the supplementary material) that models the distribution of rounding drift to conservatively bound  $\hat{\tau}^*$ .

**Support is not truth.** The auditor measures adherence to *retrieved* evidence. If the corpus is false, stale, biased, or poisoned, a faithful generation can still propagate harm. Source authentication, retrieval evaluation, and human oversight remain separate requirements.

**Whole-answer dilution.** Mean pooling produces a routing score, not a localization guarantee. RAGTruth makes this limitation visible: a short unsupported span can disappear inside an otherwise supported response. Claim-scoped scoring works in our controlled case study, but that study does not replace a natural claim-level benchmark. LatentAudit should not by itself suppress, approve, or revise individual claims.

**Verification scope.** The zero-knowledge layer certifies that the reported audit score was computed correctly; it says nothing about the quality of the latent signal. The proof system is a cryptographic convenience, not a substitute for the empirical ML validation presented above.

**Calibration and population shift.** The affine projector, covariance, and threshold are estimated from labeled calibration data. They must be revalidated when the generator, retrieval stack, language, or user population changes. The 32B experiment extends the original scale range, but 70B+ and closed frontier models remain untested.

**Misuse and disparate impact.** A risk score can create false reassurance or route disproportionate review burden to particular topics or writing styles. Deployers should report false-negative and false-positive rates by domain and user group, preserve an appeal path, and avoid treating the score as evidence of intent or competence. The optional proof layer attests computation, not fairness or correctness.

## 6 Conclusion

LatentAudit tests a simple proposition: faithful answer states leave a repeatable residual pattern after alignment with retrieved evidence. Across the tested domains and open-weight models, that pattern supports a fast local warning without a second generative judge. RAGTruth supplies the equally important boundary: response pooling can miss localized unsupported content. The resulting operating point is narrow but useful. LatentAudit can route answers for review when compute and data egress matter, provided the generator is open, retrieval is trusted, and calibration is revalidated after shifts. It is one monitoring component, not a complete safety case.

## References

- Amos Azaria and Tom Mitchell. The internal state of an LLM knows when it’s lying. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 967–976, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.68. URL <https://aclanthology.org/2023.findings-emnlp.68/>.
- Chao Chen, Kai Liu, Ze Chen, Yi Gu, Yue Wu, Mingyuan Tao, Zhihang Fu, and Jieping Ye. INSIDE: llms’ internal states retain the power of hallucination detection. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024. URL <https://openreview.net/forum?id=Zj12nz1Qbz>.
- Jens Groth. *On the Size of Pairing-Based Non-interactive Arguments*, pages 305–326. Springer Berlin Heidelberg, 2016. ISBN 9783662498965. doi: 10.1007/978-3-662-49896-5\_11. URL [http://dx.doi.org/10.1007/978-3-662-49896-5\\_11](http://dx.doi.org/10.1007/978-3-662-49896-5_11).
- Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, L  lio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timoth  e Lacroix, and William El Sayed. Mistral 7b, 2023. URL <https://arxiv.org/abs/2310.06825>.
- Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William Cohen, and Xinghua Lu. PubMedQA: A dataset for biomedical research question answering. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan, editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2567–2577, Hong Kong, China, November 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1259. URL <https://aclanthology.org/D19-1259/>.
- Mandar Joshi, Eunsol Choi, Daniel Weld, and Luke Zettlemoyer. TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension. In Regina Barzilay and Min-Yen Kan, editors, *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1601–1611, Vancouver, Canada, July 2017. Association for Computational Linguistics. doi: 10.18653/v1/P17-1147. URL <https://aclanthology.org/P17-1147/>.
- Junyi Li, Xiaoxue Cheng, Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. HaluEval: A large-scale hallucination evaluation benchmark for large language models. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6449–6464, Singapore, December 2023a. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.397. URL <https://aclanthology.org/2023.emnlp-main.397/>.
- Kenneth Li, Oam Patel, Fernanda Vi  gas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 41451–41530. Curran Associates, Inc., 2023b. doi: 10.52202/075280-1797. URL [https://proceedings.neurips.cc/paper\\_files/paper/2023/file/81b8390039b7302c909cb769f8b6cd93-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2023/file/81b8390039b7302c909cb769f8b6cd93-Paper-Conference.pdf).
- Potsawee Manakul, Adian Liusie, and Mark Gales. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9004–9017, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.557. URL <https://aclanthology.org/2023.emnlp-main.557/>.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in gpt. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, editors, *Advances in Neural Information Processing Systems*, volume 35, pages 17359–17372. Curran Associates, Inc., 2022. doi: 10.52202/

- 068431-1262. URL [https://proceedings.neurips.cc/paper\\_files/paper/2022/file/6f1d43d5a82a37e89b0665b33bf3a182-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2022/file/6f1d43d5a82a37e89b0665b33bf3a182-Paper-Conference.pdf).
- Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, KaShun Shum, Randy Zhong, Juntong Song, and Tong Zhang. RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10862–10878, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.585. URL <https://aclanthology.org/2024.acl-long.585/>.
- Llama Team. The llama 3 herd of models. *CoRR*, abs/2407.21783, 2024. doi: 10.48550/ARXIV.2407.21783. URL <https://doi.org/10.48550/arXiv.2407.21783>.
- Qwen Team. Qwen3 technical report. *CoRR*, abs/2505.09388, 2025. doi: 10.48550/ARXIV.2505.09388. URL <https://doi.org/10.48550/arXiv.2505.09388>.
- Di Wu, Jia-Chen Gu, Fan Yin, Nanyun Peng, and Kai-Wei Chang. Synchronous faithfulness monitoring for trustworthy retrieval-augmented generation. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen, editors, *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 9390–9406, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.527. URL <https://aclanthology.org/2024.emnlp-main.527/>.
- An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report. *CoRR*, abs/2412.15115, 2024. doi: 10.48550/ARXIV.2412.15115. URL <https://doi.org/10.48550/arXiv.2412.15115>.
- Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In Ellen Riloff, David Chiang, Julia Hockenmaier, and Jun’ichi Tsujii, editors, *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380, Brussels, Belgium, October–November 2018. Association for Computational Linguistics. doi: 10.18653/v1/D18-1259. URL <https://aclanthology.org/D18-1259/>.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, Hao Zhang, Joseph Gonzalez, and Ion Stoica. Judging llm-as-a-judge with mt-bench and chatbot arena. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 46595–46623. Curran Associates, Inc., 2023. doi: 10.52202/075280-2020. URL [https://proceedings.neurips.cc/paper\\_files/paper/2023/file/91f18a1287b398d378ef22505bf41832-Paper-Datasets\\_and\\_Benchmarks.pdf](https://proceedings.neurips.cc/paper_files/paper/2023/file/91f18a1287b398d378ef22505bf41832-Paper-Datasets_and_Benchmarks.pdf).
- zkonduit. EZKL. GitHub repository: zkonduit/ezkl, 2022. URL <https://github.com/zkonduit/ezkl>.

## 363 A Dataset Construction Details

- 364 **PubMedQA.** We use the expert-labeled (PQA-L) split of PubMedQA [Jin et al., 2019], which  
 365 contains 1,000 question–answer pairs with biomedical abstracts as evidence. We retain the yes/no-  
 366 labeled instances and exclude the “maybe” class, yielding 800 filtered seeds. Evidence text is the  
 367 concatenation of all labeled abstract sections. For the stress test, retrieval-miss examples replace the  
 368 original evidence with topically similar but source-mismatched biomedical snippets, while partial  
 369 examples retain only weak or incomplete sections from the original abstract.
- 370 **TriviaQA.** We sample 800 evidence-grounded instances from the web-verified split of TriviaQA  
 371 [Joshi et al., 2017]. Evidence paragraphs are truncated to 512 tokens. We generate contradicted

372 variants by entity-swapping the gold answer with a same-type distractor drawn from the same  
 373 evidence paragraph.

374 **HotpotQA.** We draw 800 multi-hop bridge questions from the distractor setting of HotpotQA [Yang  
 375 et al., 2018]. Evidence is the concatenation of the two gold supporting paragraphs. Partial evidence  
 376 removes one of the two supporting documents, forcing single-hop reasoning.

377 **Stress-test expansion.** For each seed, `build_paper_stress_eval.py` generates four evaluation  
 378 records: (i) *faithful* (original evidence), (ii) *contradicted* (entity-swapped answer), (iii) *retrieval-miss*  
 379 (topically similar but source-mismatched evidence), and (iv) *partial* (evidence with key supporting  
 380 spans removed). Embedding-space diversity is enforced by selecting retrieval-miss candidates via  
 381 farthest-point sampling in a 768-dimensional sentence-embedding space.

## 382 B External-Benchmark, Mixed-Claim, and Scale Evaluation Details

Table 10: Incremental audit profile for Qwen2.5-32B ( $d = 5120$ , L32, bfloat16) on one A100. Shared generation and the cached evidence embedding are shown for context but are not included in the 0.24 ms audit total.

| Component                      | Latency (ms) |
|--------------------------------|--------------|
| Token-vector extraction        | 0.09         |
| Activation mean pooling        | 0.05         |
| Evidence projection            | 0.02         |
| Mahalanobis quadratic form     | 0.05         |
| Cosine and threshold           | 0.03         |
| <b>Incremental audit total</b> | <b>0.24</b>  |
| RAG forward pass (shared)      | ~111         |
| Evidence embedding (cached)    | ~106         |

383 The RAGTruth values  $0.7816$  and  $0.788 \pm 0.001$  use different protocols. The first uses 200-sample  
 384 closed-form recalibration; the second uses a repeated-split six-feature bundle. The artifact release  
 385 records separate split manifests, calibration objects, and predictions for each protocol.

Table 11: Attention-weighted pooling check (Qwen2.5-7B, L14, 800 PubMedQA stress examples, five seeds). This protocol differs from the full 1,600-example pooling ablation reported later.

| Pooling            | Mean AUROC | Range         |
|--------------------|------------|---------------|
| Last token         | 0.6814     | 0.6381–0.7362 |
| Mean               | 0.6414     | 0.5827–0.6950 |
| Attention-weighted | 0.6330     | 0.5798–0.6777 |

## 386 C Hyperparameter Summary

387 **Salient token selection.** The auditor selects the top- $k$  answer tokens by TF-IDF, with inverse  
 388 document frequencies computed on the calibration corpus. It mean-pools their activations and  
 389 applies the Mahalanobis rule. Appendix I compares this centroid with last-token and max-pooling  
 390 alternatives.

391 **Threshold calibration.** We fit  $\tau^*$  as the operating point that maximizes Youden’s  $J$  on the calibra-  
 392 tion split. Over 200 bootstrap resamples,  $\tau^*$  varies by  $\sigma=0.063$  on PubMedQA and  $\sigma=0.086$  on  
 393 HotpotQA.

## 394 D Latent Geometry Distributions

395 Figure 4 disaggregates PubMedQA scores by model. Every tested family separates the faithful and  
 396 contradicted populations, although the distribution shapes differ. The aggregate panel summarizes  
 397 rather than replaces these per-model views.

Table 12: Hyperparameters and configuration choices.

| Component             | Parameter                      | Value                                   |
|-----------------------|--------------------------------|-----------------------------------------|
| Activation extraction | Target layer $L$               | 16 (Llama), 14 (Qwen), 15 (Mistral)     |
|                       | Pooling                        | Mean over salient answer tokens         |
|                       | Salient token count            | 8                                       |
| Alignment scoring     | Distance metric                | Mahalanobis ( $D_M$ )                   |
|                       | Covariance estimator           | Ledoit–Wolf shrinkage                   |
|                       | Threshold $\tau^*$             | Maximum Youden’s $J$ on calibration set |
| ZK quantization       | Fixed-point bits $k$           | 16                                      |
|                       | Field prime $p$                | BN254 scalar field                      |
| Evaluation            | Summary-table resamples        | 5                                       |
|                       | Threshold-stability bootstraps | 200                                     |
|                       | Calibration/evaluation split   | 10% / 90% stratified                    |

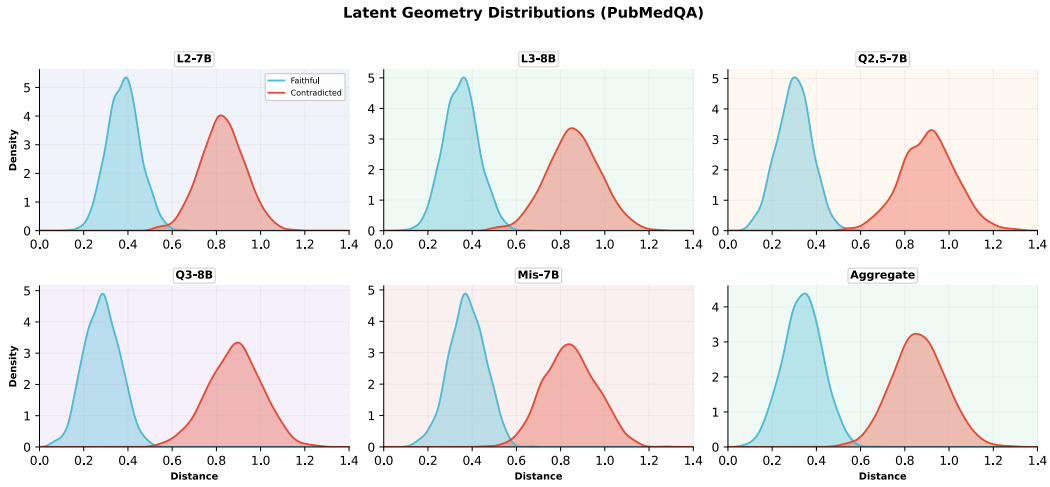


Figure 4: Per-model Mahalanobis distance distributions on PubMedQA. Each panel shows the KDE of faithful (blue) and contradicted (red) alignment scores. The bottom-right panel aggregates all models.

## E Code Architecture

The codebase is organized as a Python package (`rag_audit`) with five principal modules:

- `alignment/` — Core audit logic. `scorer.py` implements cosine similarity and centroid computation; `auditor.py` orchestrates the full audit pipeline (salient token extraction → centroid pooling → alignment scoring → threshold classification); `threshold.py` manages calibrated decision boundaries.
- `model/` — Hugging Face model loading and activation extraction. Supports Llama-2/3, Qwen-2.5/3, and Mistral families via a unified `GenerationResult` interface that captures per-token hidden states.
- `proof/` — Zero-knowledge proof pipeline. `quantizer.py` maps floating-point activations to  $\mathbb{F}_p$ ; `circuit_input.py` assembles the witness; `prover.py` generates proof artifacts; `verifier.py` validates them.
- `retrieval/` — Vector store abstraction for evidence retrieval and embedding management.
- `datasets/` — Data loaders for PubMedQA, TriviaQA, and HotpotQA.

All experiments are driven by `scripts/run_activation_audit_experiment.py`, which takes a model path and shard index and writes per-sample audit results to JSONL. The stress-test evalu-

414 ation sets are built by `scripts/build_paper_stress_eval.py`, which constructs the four-way  
 415 faithful/contradicted/retrieval-miss/partial splits described in Section 4.

## 416 F ZK Circuit Details

417 The ZK circuit verifies the inequality  $\hat{D}_M \leq \hat{\tau}^*$  in  $\mathbb{F}_p$  (BN254 scalar field). The circuit takes as  
 418 public inputs the quantized threshold  $\hat{\tau}^*$ , the trace hash, and the audit ID. The witness contains the  
 419 quantized activation centroid, evidence vector, and inverse covariance matrix.

420 The Solidity verifier (`AuditVerifier.sol`) consumes  $\sim 580.6K$  gas, dominated by the elliptic-curve  
 421 pairing precompiles (`ecPairing`). On Ethereum L1 at 30 Gwei gas price, this costs \$21.77 per  
 422 verification; on Arbitrum L2 the same call costs \$1.09. In the fixed-point experiment,  $k=16$  preserves  
 423  $>99.8\%$  of the FP16 AUROC.

## 424 G Per-Model Detailed Results

Table 13: Full per-model AUROC / F1 on PubMedQA.

| Method                    | Llama-2  | Llama-3  | Qwen-2.5 | Qwen-3   | Mistral  |
|---------------------------|----------|----------|----------|----------|----------|
| GPT-4o Judge              | .948/.87 | .948/.87 | .948/.87 | .948/.87 | .948/.87 |
| SelfCheckGPT              | .862/.80 | .871/.81 | .855/.79 | .869/.80 | .858/.79 |
| INSIDE-style <sup>†</sup> | .903/.83 | .908/.84 | .899/.82 | .905/.83 | .895/.82 |
| SAPLMA-style <sup>†</sup> | .878/.81 | .882/.82 | .872/.80 | .880/.81 | .870/.80 |
| Min-Perplexity            | .718/.69 | .722/.70 | .715/.68 | .720/.69 | .712/.68 |
| LatentAudit               | .931/.86 | .942/.87 | .938/.86 | .948/.88 | .925/.85 |

425 Rankings are consistent across the five tested families. LatentAudit exceeds the controlled INSIDE-  
 426 and SAPLMA-style adaptations and remains below or level with GPT-4o. This supports transfer  
 427 within the evaluated architectures, not beyond them.

## 428 H Extended Deployment Analysis

429 Figure 5 expands the RQ3 cost analysis. Panel (a) reports measured batching throughput for the  
 430 quadratic score and autoregressive baselines. Panel (b) places their measured quality and cost on a  
 431 common Pareto plot.

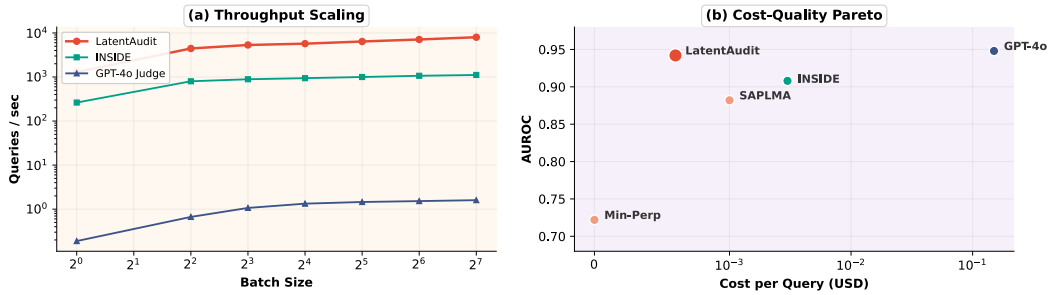


Figure 5: Extended deployment analysis. (a) Throughput scaling with batch size. (b) Cost-quality Pareto front.

## 432 I Methodology Ablations

433 Table 14 varies pooling and the number of TF-IDF-selected tokens. Mean-pooling with  $k \in [4, 16]$   
 434 exceeds last-token and max-pooling in reported AUROC. Performance peaks at  $k = 8$  and declines  
 435 slightly as less salient tokens enter the centroid.

436 Reducing the calibration share from 10% to 5% lowers PubMedQA AUROC by 0.003 on average.  
 437 Within this range, the fitted rule is not highly sensitive to calibration size.

Table 14: PubMedQA AUROC across pooling strategies and token counts in Llama-3-8B.

| Pooling Strategy | $k = 1$ (Last) | $k = 4$ | $k = 8$      | $k = 16$ | $k = 32$ |
|------------------|----------------|---------|--------------|----------|----------|
| Mean-Pool        | 0.884          | 0.933   | <b>0.942</b> | 0.940    | 0.931    |
| Max-Pool         | 0.884          | 0.901   | 0.912        | 0.908    | 0.895    |

## 438 J Continuous Safety Margin for Quantization

439 Quantization can change decisions for samples near the boundary of the  $\mathbb{F}_p$  circuit. We therefore  
 440 introduce a safety margin  $\varepsilon(k)$  at fractional precision  $k$ . The corresponding bound on drift in the  
 441 quadratic form is  $\varepsilon(k) = \mathcal{O}(d \cdot 2^{-k} \cdot \lambda_{\max}(\Sigma^{-1}))$ .

442 We configure the on-chain threshold conservatively as  $\hat{\tau}_{\text{safe}}^* = \hat{\tau}^* - \varepsilon(k)$ . At  $k = 16$ , the largest  
 443 observed score drift is approximately 0.04. No labeled unsupported example in the measured set  
 444 crosses the adjusted boundary; this is not a guarantee outside that set.

## 445 K Robustness of the Affine Projector $W_{\text{proj}}$

446 LatentAudit maps an external evidence embedding into the generator’s answer-state space. We test  
 447 how much performance comes from this supervised alignment by varying projector complexity and  
 448 calibration size on Llama-3-8B and PubMedQA.

449 Table 2 in the main paper compares projection strategies. Unsupervised PCA alignment reaches 0.778  
 450 evaluation AUROC, while Ridge reaches 0.942. A two-layer MLP reaches a similar 0.945 but has a  
 451 much larger train–evaluation gap (0.991 versus 0.945). We retain Ridge because it is closed-form,  
 452 simpler, and nearly matches the nonlinear alternative; we do not claim that Ridge has the highest  
 453 point estimate.

454 Table 8(a) varies the number of faithful calibration pairs. Evaluation AUROC rises from 0.912 at 50  
 455 pairs to 0.942 at 200, then plateaus at 0.943. The plateau supports the 200-pair default, but it does not  
 456 by itself identify what representation the projector learns.